Abishek Prakash Nagaram

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Summary

AI engineer and M.S. Computer Science student at **Arizona State University** (GPA 3.72/4.0), graduating May 2027, who builds production AI features and trains computer vision models. At **Selfmadee.ai**, I shipped LLM and full stack features for a multi tenant SaaS platform and planned a move to open source models projected to cut AI spend by **64%**. At **Jireh Software Solutions**, I built an OCR and GPT-4 tool that cut my manager's daily review time from **1.5 hours to 10 minutes**. In my ASU research, I train medical imaging models on **A100 GPUs** and **Intel Gaudi** accelerators and beat the published **MedMNIST v2** baselines on two 3D datasets (**0.937 vs 0.907 AUC** on VesselMNIST3D and **0.872 vs 0.828** on NoduleMNIST3D).

Education

Arizona State University **Tempe, AZ**

Master's in Computer Science (GPA: 3.72/4.0, May 2027) Foundation Models & Medical Imaging, Agentic AI, Statistical ML

Dayananda Sagar University **Bangalore, India**

Bachelor's in Computer Science (Jun 2025)

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
ML & Computer Vision: PyTorch, TensorFlow, Vision Transformers (ViT, Swin), CNNs, Model Training & Evaluation
Generative AI & Agents: Agentic AI, Multi-Agent Systems, LLM APIs (OpenAI, Claude, Gemini), RAG, Embeddings, Vector Databases (pgvector), LangChain, vLLM, Hugging Face, Prompt Engineering
Backend & Web: FastAPI, Django, Node.js, React, REST APIs, Supabase, Streamlit
Data & Infrastructure: PostgreSQL, MongoDB, pandas, Docker, AWS Bedrock, Linux, Git, SLURM (A100 and Intel Gaudi clusters)

Work Experience

Selfmadee.ai **Remote**

AI Engineering Intern *Jun 2026 – Aug 2026*

- Audited all **43 AI features** across the platform's **9 product modules** (such as HR, Finance, Marketing, and Supply Chain), mapping each one's code path, model, usage, and trigger, and recommended keeping 32, merging 4, deprecating 3, and removing 4 to cut redundant AI calls.
- Compared **AWS Bedrock Llama** models with the GPT models in use on price, context length, and task fit, and proposed a 3-phase switch projected to cut monthly AI cost **64% ($329 to $119 at 1,000 chats a day)**.
- Built features with an **AI coding agent (Claude Code)**, reviewing its output for bugs and security issues, and set up a **searchable knowledge graph of the 200+ function codebase** for the team and AI agents.
- Shipped an **Instagram and Facebook analytics dashboard** to production (**React, TypeScript, Supabase, Meta Graph API**, +1.9K lines) showing engagement trends, top posts, and the best times to post.
- Found a **data-isolation flaw** that could expose one customer's records to another (missing company ID on a table with an open access policy), wrote the fix plan, and fixed a purchase-order bug that skipped payment capture.

Jireh Software Solutions **Bangalore, India**

Python Developer Intern *Feb 2025 – May 2025*

- Built an **employee activity analyzer** solo in about 6 weeks that read ~**320 work screenshots a day across 8 employees** and sorted each person's time into **9 activity types** (coding, meetings, research, idle, and more), using **EasyOCR** for text extraction and the **GPT-4 API** for classification.
- Cut my manager's daily review time from ~**1.5 hours to ~10 minutes** by scoring each employee's efficiency against their to-do list and daily Google Sheets logs (**pandas**). He used its output for his **reports to the CTO**, and the tool stayed in **daily use after handoff**.
- Delivered results through a **Streamlit chat app** with per-employee **Matplotlib** efficiency charts, and improved accuracy by removing duplicate screenshots and flagging fullscreen and mid-transition captures.
- Compared **open-source chat LLMs on Hugging Face** (Meta, Google, and Qwen models) for a client's art e-commerce marketplace and deployed **Qwen2.5-7B** with **vLLM**, handling prompt engineering and QA testing.

Projects

Chest X-ray Foundation Models & Benchmarking | *PyTorch, Swin, ViT, SLURM, Intel Gaudi* **ASU Research**

- Pretrained **Ark+**, a chest X-ray foundation model (**Swin-B** with teacher-student distillation), on **112K ChestX-ray14 images** using SLURM jobs on ASU's Sol cluster (**A100 GPUs and Intel Gaudi accelerators**).
- Replicated 3 **CycliST** experiments from a 2026 thesis on **Intel Gaudi2** (one head per radiologist across 17 radiologists, a noisy 18th annotator, and 4 imbalanced hospital sites), showing joint per-site training beats isolated specialists by **+4.5 AUC (+5.2 on other diseases)**, and traced a 7 to 8 point gap to a likely **learning-rate scheduler bug** in Ark+.
- Trained **Foundation X+** to classify, locate, and segment collapsed lungs (pneumothorax) in a single model (**DINO + UperNet**) on CANDID-PTX, testing all 7 task combinations and reaching **87.6 Dice, 76.8 AUC, and 43.9 mAP@50**.
- Reproduced **MedMNIST v2** benchmarks with 2D/3D ResNet-50 (5 runs per dataset), matching OrganMNIST3D (**0.993 AUC**) and beating the published baselines on NoduleMNIST3D (**0.872 vs 0.828**) and VesselMNIST3D (**0.937 vs 0.907**).

Tau-Bench Multi-Agent Tool-Use Agent | *Python, vLLM, Qwen3, SLURM, Gaudi* **ASU Team Project, Spring 2026**

- Built a **planner, executor, and validator multi-agent system** for Tau-Bench, a benchmark where an agent chats with a simulated customer and makes **irreversible database changes under airline policy**. Using the same **Qwen3-Coder-30B** model as the tool-calling baseline, it raised `pass^1` **from 28% to 36%** and `pass^3` **from 14% to 26%** on the Tau-Bench airline domain.
- Replaced a hallucinating LLM critic with a **deterministic validator** that blocks made-up user and order IDs before any write, at **zero extra LLM calls**, added a **policy gate** that holds irreversible actions until conflicting evidence is resolved, and fixed an orchestrator loop that was using up the 30-step limit.
- Benchmarked **Act, ReAct, and function-calling** agents across **Qwen3 4B to 32B** on the Tau-Bench airline domain, and built an **error taxonomy from 150+ failure cases** (hallucinated IDs, scope drift, policy violations, wrong tool for an order's state) that shaped the multi-agent design.